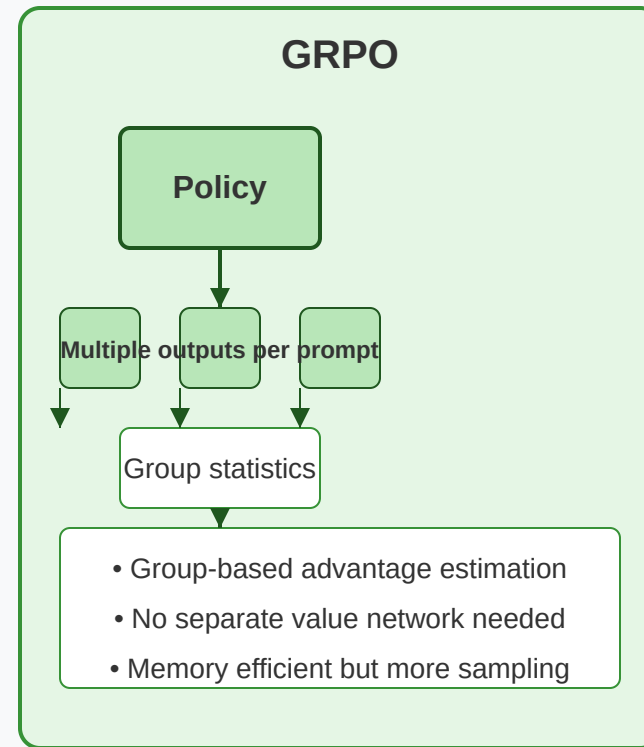
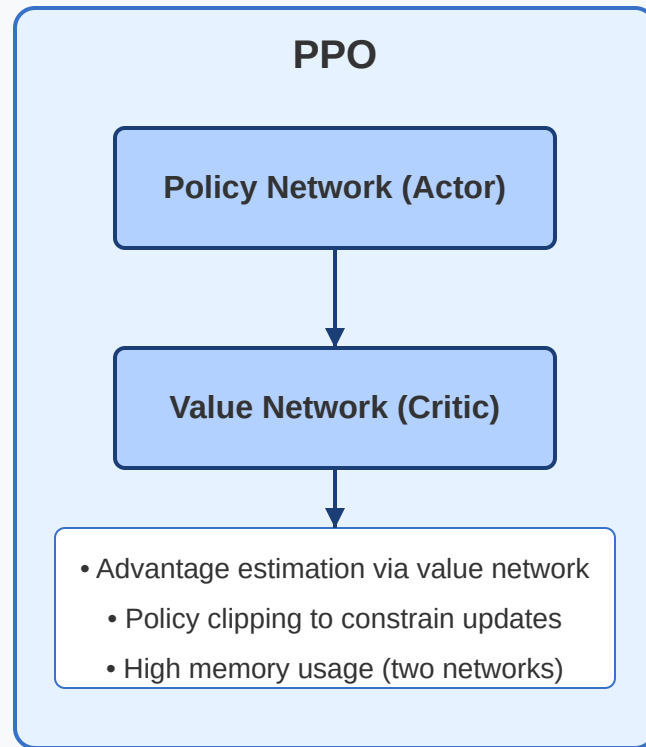


Enhancing LLM Fine-tuning with Knowledge Distillation

DA-PPO, DA-GRPO, and HRLD Approaches

PPO vs GRPO Architecture Comparison

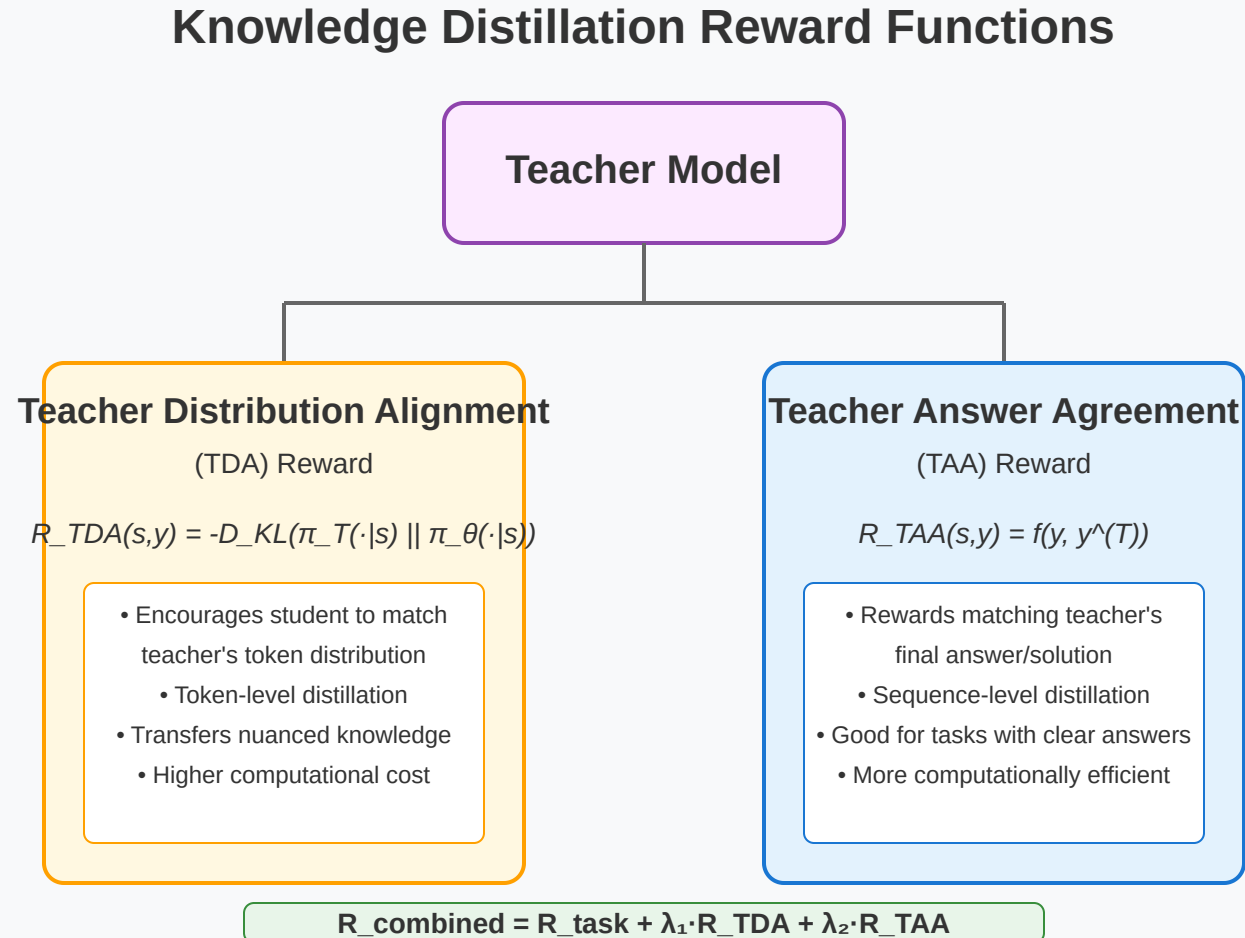
PPO vs GRPO Architecture Comparison



PPO vs GRPO: Summary

- This graphic illustrates the fundamental architectural differences between PPO and GRPO for LLM fine-tuning:
- **PPO (Proximal Policy Optimization):**
 - Uses two separate networks: policy (actor) and value (critic).
 - Value network estimates advantages for policy updates.
 - Requires significant memory but provides stable learning.
- **GRPO (Group Relative Policy Optimization):**
 - Eliminates the separate value network, using only a policy network.
 - Generates multiple outputs (group sampling) per prompt.
 - Computes advantages based on relative rewards within the group (mean/std).
 - More memory efficient, but requires more sampling per step.
- **Key Innovation:** GRPO replaces the critic network with group-based advantage calculation, suitable for large models where duplicating the network is costly.

Knowledge Distillation Reward Functions

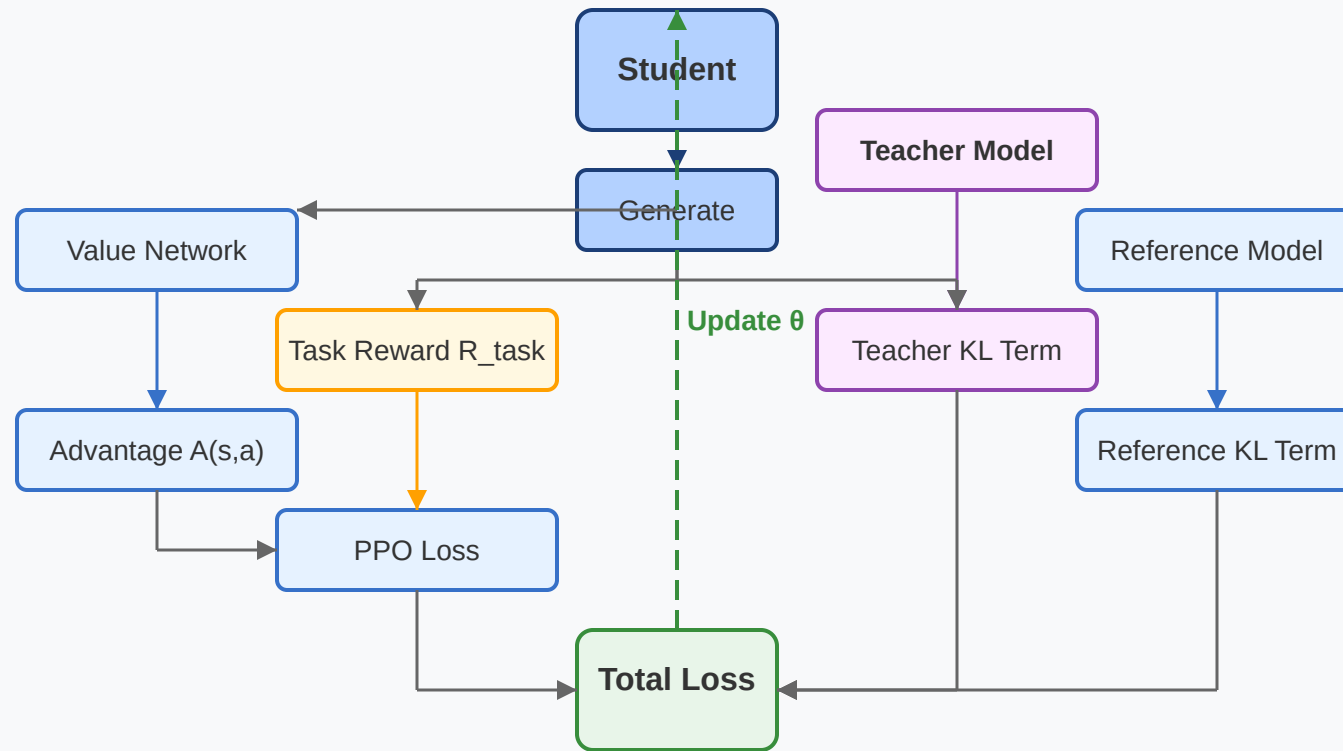


Knowledge Distillation Rewards: Summary

- Two novel reward functions proposed for integrating teacher knowledge:
- **Teacher Distribution Alignment (TDA) Reward:**
 - Measures divergence between student and teacher output distributions (KL divergence).
 - Acts at the token level: $R_{TDA} = -D_{KL}(\pi_T || \pi_\theta)$
 - Provides fine-grained guidance, transferring nuanced knowledge (style, token choices).
 - Higher computational cost.
- **Teacher Answer Agreement (TAA) Reward:**
 - Measures similarity between the student's final output and the teacher's.
 - Acts at the sequence level: $R_{TAA} = f(y, \hat{y}(T))$ (similarity function).
 - More efficient computationally, ideal for tasks with clear correct answers.
 - Focuses on end results.
- **Combined Reward:** Can be combined with task rewards: $R_{combined} = R_{task} + \lambda_1 \cdot R_{TDA} + \lambda_2 \cdot R_{TAA}$

Distillation-Augmented PPO (DA-PPO)

Distillation-Augmented PPO (DA-PPO) Algorithm

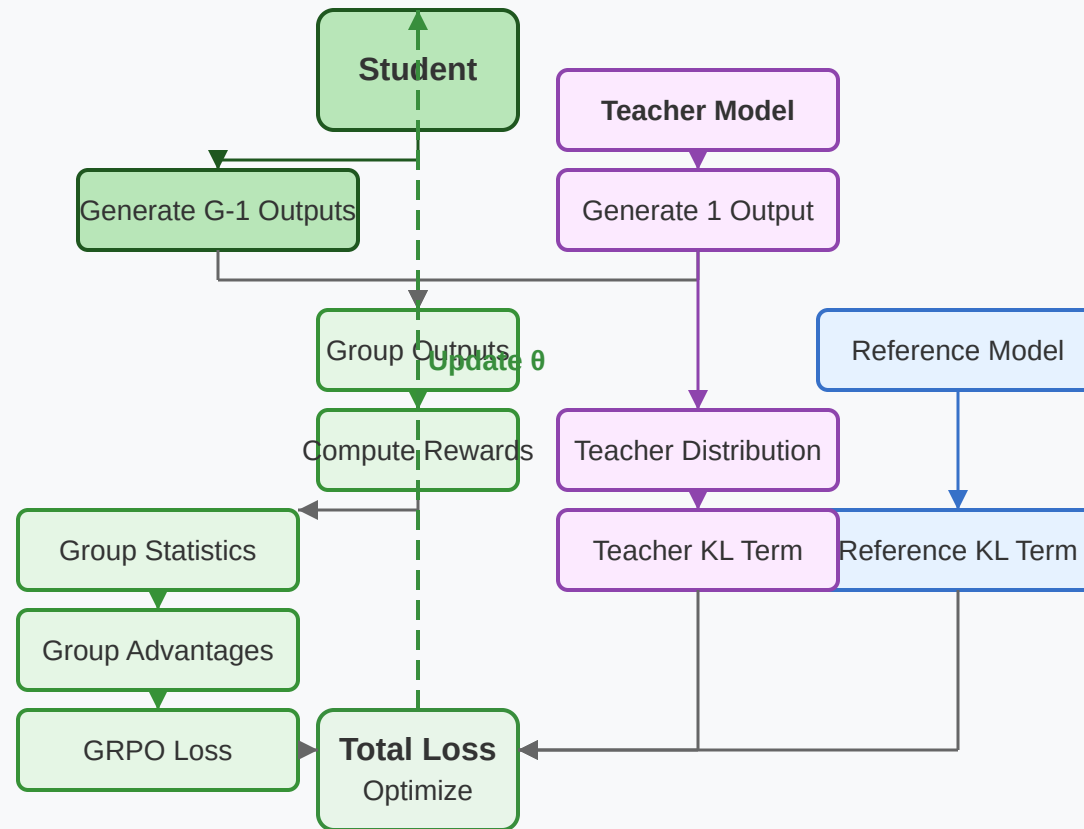


DA-PPO Algorithm: Summary

- Illustrates how DA-PPO enhances standard PPO with knowledge distillation:
- **Key Components:** Student Model, Teacher Model, Reference Model, Value Network (Critic).
- **Process Flow:**
 1. Student generates outputs and receives task rewards.
 2. Teacher provides distributions for knowledge guidance.
 3. KL divergence between student and teacher is calculated (Teacher KL Term).
 4. Value network estimates advantages for PPO updates.
 5. Total loss combines: PPO loss, Teacher KL penalty, Reference KL penalty.
 6. Student parameters updated to optimize combined loss.
- Effectively blends RL with KD, learning from rewards and teacher guidance simultaneously.

Distillation-Augmented GRPO (DA-GRPO)

Distillation-Augmented GRPO (DA-GRPO) Algorithm

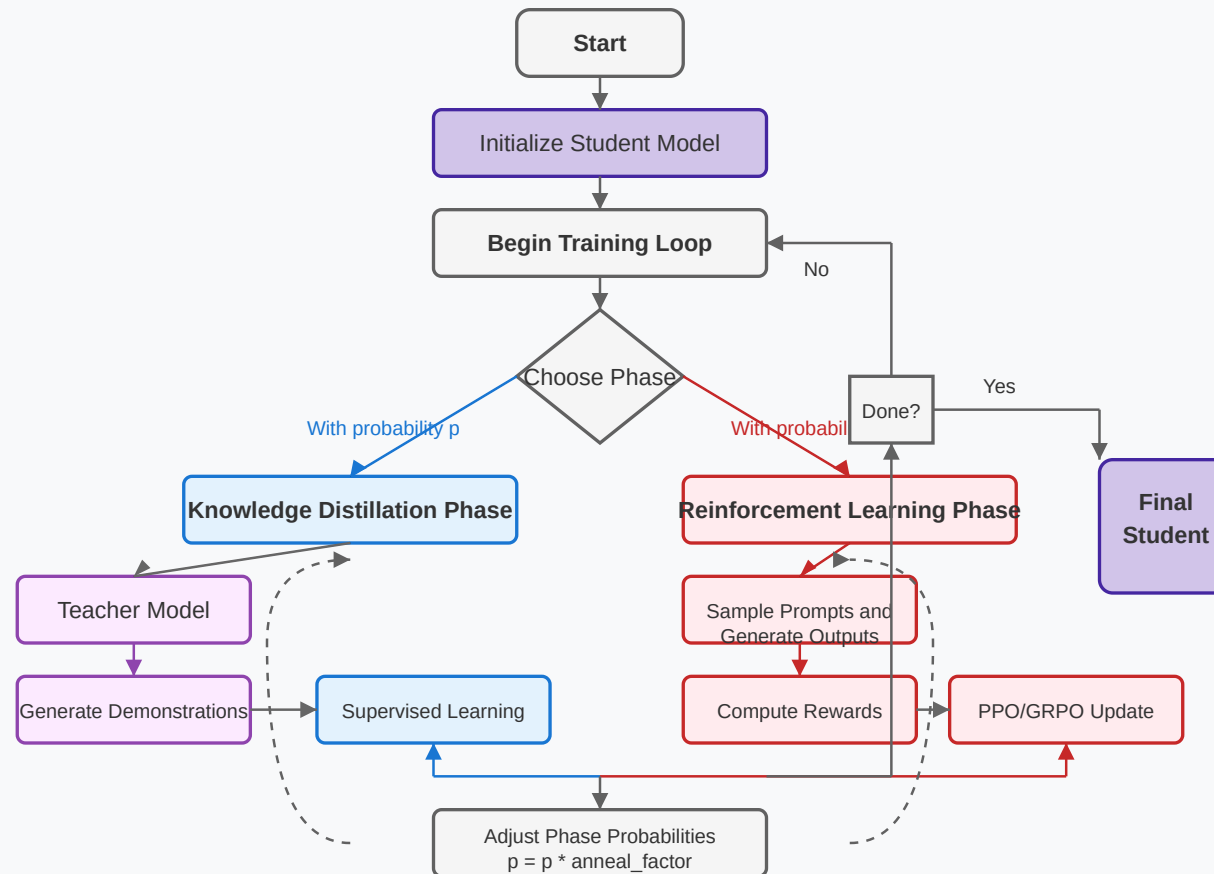


DA-GRPO Algorithm: Summary

- Shows how DA-GRPO integrates teacher knowledge into the GRPO framework:
- **Key Components:** Student Model, Teacher Model, Group Outputs (incl. teacher's), Reference Model.
- **Process Flow:**
 1. Student generates G-1 outputs, Teacher generates 1 output per prompt.
 2. These G outputs form a response group.
 3. Rewards computed for all outputs (including teacher's).
 4. Group statistics (mean, std) calculated for reward normalization.
 5. Relative advantages computed using normalized reward differences.
 6. Teacher provides token distributions for KL calculation.
 7. Total loss combines: GRPO loss, Teacher KL term, Reference KL term.
 8. Student parameters updated.
- **Key Innovation:** Including the teacher's output directly in the group baseline calculation forces the student to compete with/match the teacher's performance.

Hybrid RL and Distillation (HRLD)

Hybrid Reinforcement Learning and Distillation (HRLD) Framework



HRLD Framework: Summary

- Presents HRLD's interleaved two-phase training approach:
- **Key Components:** Training Loop, Phase Selection logic, KD Phase, RL Phase, Annealing Schedule.
- **Process Flow:**
 1. Initialize student model.
 2. Begin training loop, dynamically choosing phase based on probability p .
 3. **Distillation Phase:** Teacher generates demonstrations; student updated via supervised learning to match.
 4. **Reinforcement Learning Phase:** Student generates outputs, receives rewards; PPO/GRPO update performed.
 5. Adjust phase probability p (annealing schedule, typically reducing distillation probability over time).
 6. Continue until training complete.
- Allows rapid knowledge absorption followed by gradual fine-tuning to reward structure, balancing knowledge transfer and alignment.

Multi-Domain Evaluation Framework

- Illustrates the comprehensive evaluation approach across seven diverse domains:
- **Creative Writing:** Human preference, coherence, originality.
- **Factual QA:** Exact match, F1 score, hallucination rate.
- **Logical Reasoning:** Answer accuracy, step-by-step correctness.
- **Code Generation:** Pass@k metrics, functional correctness.
- **Dialogue:** Engagement, coherence, human preference.
- **Instruction Following:** Compliance rate, task-specific metrics.
- **Domain Adaptation:** Zero-shot performance, fine-tuning efficiency.
- Statistical significance testing applied across domains for robust results.
- Ensures algorithms generalize well beyond narrow tasks.

Conclusion

- DA-PPO, DA-GRPO, and HRLD represent a comprehensive suite of approaches for enhancing LLM fine-tuning with knowledge distillation.
- By leveraging teacher model expertise (via distribution matching, answer agreement, or direct inclusion in training), they aim to produce more capable student models.
- The goal is to maintain performance across multiple domains while optimizing for human preferences or task-specific rewards, effectively transferring knowledge from a stronger teacher model.
- The multi-domain evaluation framework ensures rigorous testing of generalization capabilities.